

## Stereo Vision

- Compute depth from two views
- Many things left unsaid: what is known, and what is unknown



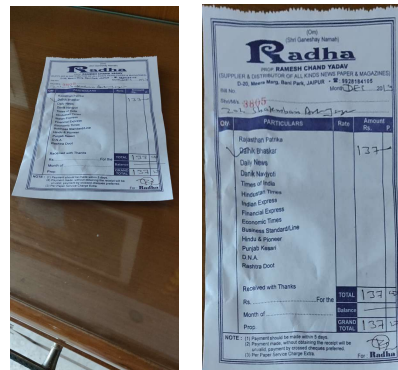
1

## Single View

Nothing ==  
casual picture

- Given a single view picture
  - If we know “nothing” about the camera
- We can still do useful things

**Nothing:** the camera is uncalibrated, i.e., we know neither the intrinsic, nor the extrinsic parameters



2

## Single View

- Given a picture,
  - If we know nothing about the camera,
  - We know *something* about the scene
- We can do useful things: single view metrology

Something: Two known distances



3

## Single View

- Given a picture,
  - If the camera intrinsics are known
  - we know something about the scene
- We can do useful things



Distance of camera from person

4

## Single View

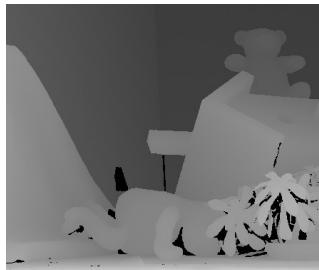
- Given a picture, what can't we do?
  - Cannot declare (longitude, latitude)
  - Cannot measure angles, length
  - Cannot measure absolute distance from camera



5

## Stereo Vision

- Compute depth from two views
- When we are given **casual pictures**, not necessarily from a device such as a Kinect or *calibrated* cameras



6

## Overview

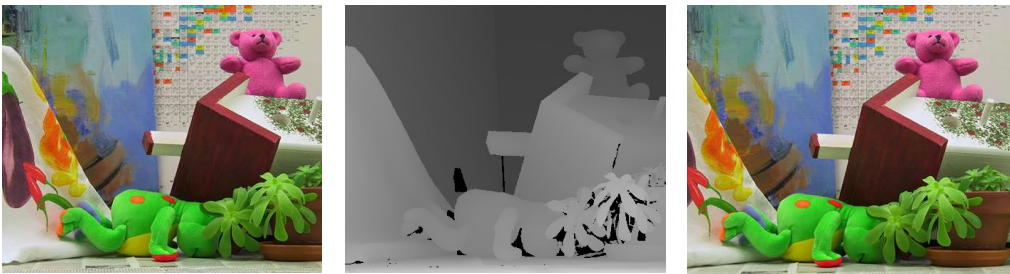
- How to do reconstruction if 'parameters' are known
- Some fun stuff!!
- Basics, buzzwords, and terminology on stereo
- The fundamental matrix  $F$ 
  - Computing 'parameters' from  $F$
- The essential matrix  $E$ 
  - Computing 'parameters' from  $E$

7

## Stereo Vision: Known cameras

- Compute depth from two views

Not Casual: Stereo parameters ( $R, T$ ) are known with 100% accuracy



8

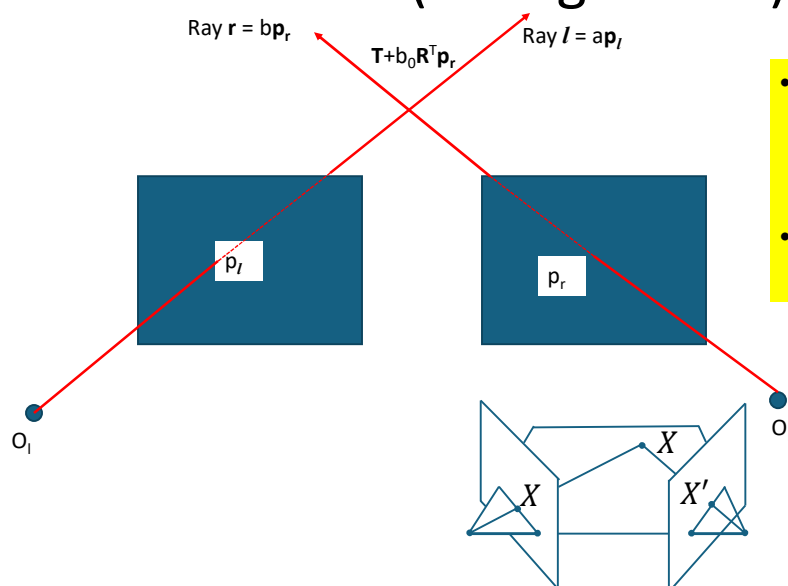
## Parameters of a stereo system

- Intrinsic parameters – focal lengths, optical centers, camera resolutions (may be same)
  - Transformations mapping a 3D point to pixel coordinates in each camera
- Extrinsic parameters – rotation  $R$  and translation  $T$  that align the coordinate systems of the two cameras
  - Transformations that align the relative position and orientation of the two cameras

$$X_{3 \times 1} = K_{3 \times 3} [I_{3 \times 3} \mid 0] X_{\text{cam}} \quad K = \begin{bmatrix} f & & p_x \\ & f & p_y \\ & & 1 \end{bmatrix}$$

9

## 3D Reconstruction (Triangulation)



10